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RESEARCH

Financial sentiment analysis with FUNNEL: filtered UNion for NER-based ensemble labeling

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Abstract

This paper introduces FUNNEL (Filtered UNion for NER-based Ensemble Labeling), a novel ensemble-based framework for labeling financial news that enhances the reliability of stock-specific sentiment signals. The framework integrates weak keyword-based heuristics with a transformer-based named entity recognition model (spaCy) through a weighted voting scheme, balancing precision and recall. Manual evaluation of 1,400 article–label pairs demonstrates that FUNNEL outperforms the original FNSPID labels in both accuracy and coverage. Applied across seven major companies, the framework reveals systematic differences in sentiment signals produced by FinBERT, RoBERTa, and VADER. These results indicate that integrating different labeling strategies enhances dataset reliability, coverage, and stability, providing a scalable framework for financial sentiment analysis.

Keywords Sentiment analysis · FUNNEL · Labeling · Transformer models · Deep learning · Natural language processing

JEL Classification G17 · C53 · C45 · C81

1 Introduction

Human emotion influences decisions across all areas of life; for example, whether in the creativity of a chef or the expression of a painter. In the financial world, emotions similarly shape decision-making, often surfacing in reaction to uncertainty, volatility, or unexpected events. News headlines, in particular, often carry emotionally charged or attention-grabbing language, sometimes referred to as “clickbait", designed to provoke reactions. Consequently, sentiment embedded in financial news

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has emerged as a powerful factor influencing short-term market movements, which leading to surges in buying or panic-induced sell-offs. Therefore, the study of emotion or sentiment in financial news in relation to financial markets can be of great interest and also something that has come to be much more common to include.

In recent years, sentiment analysis has gained traction in financial research, with studies showing that emotional signals embedded in news can affect asset prices, trading volumes, and volatility (Baker and Wurgler 2007; Loughran and McDonald 2011). Traditional models based on economic indicators are often slow to respond to market sentiment shifts, whereas natural language processing (NLP)[1] offers a robust (Du et al. 2025) alternative by extracting meaningful patterns from unstructured text data. This has led to growing interest in combining textual sentiment signals with numerical time series models for stock prediction. Vaswani et al. (2017) introduced the Transformer architecture, which has positively impacted a variety of NLP tasks such as sentiment analysis, machine translation, and summarization of language. This methodology of using Transformer-based models thus provides a new opportunity to be implemented in financial NLP and forecasting tasks. The performance of these models has been consistently proven in previous research to improve the capacity and predictive power of previous state-of-the-art (SOTA) machine learning models. However, these types of architecture are often deemed as data hungry compared to previous machine learning models such as recurrent neural networks (RNN) or long short-term memory (LSTM) while also being computationally intensive concerning both time and space complexity (Brown et al. 2020).

Advances in NLP, especially transformer-based models such as bidirectional encoder representations from transformers (BERT), financial text mining BERT (FinBERT) and robustly optimized BERT pre-training approach (RoBERTa) have significantly improved the accuracy of tasks like sentiment analysis and named entity recognition (NER). These models allow researchers to move beyond simple keyword-based methods and instead capture nuanced, context-aware sentiment from financial news. NLP explores the automation of language processing and generation. Its applications range from simple tasks like string matching and tokenization to advanced systems, such as ChatGPT, that use neural networks to summarize content or translate between languages.

1.1 Related works

Financial modeling gained traction, particularly with the introduction of the efficient market hypothesis (Fama 1970). While event-based modeling has been explored for decades, advances in computational power have enabled the adoption of machine

[1] NLP is a branch of Artificial Intelligence (AI) that uses machine learning to interpret, understand, and generate human language, but also predict sentiment.

learning approaches for stock price prediction, ranging from traditional statistical approaches like support vector machines (SVM) to more complex neural networks. The literature review is built in a context that forms a foundation for informed decisions, particularly regarding data modeling.

NLP has become an increasingly important tool in financial research, especially for extracting sentiment from text data. The field of NLP started in the 1950s as part of AI and has grown from basic rule-based systems to modern deep learning models capable of handling complex language tasks (Zhang and Teng (2021), pp. 3–4). One of the earliest studies linking sentiment to stock market prediction was by Bollen et al. (2011) demonstrated that public mood measured from Twitter could help predict changes in the Dow Jones industrial average, achieving nearly 88% accuracy using mood indicators and machine learning models. Simultaneously, Schumaker and Chen (2009) used financial news articles with NLP features like bag-of-words and named entities to forecast stock prices, finding that combining news with price data improved both prediction accuracy and trading returns. However, these early approaches used generic language tools, which can be misleading in finance. Loughran and McDonald (2011) showed that many common negative words in general sentiment dictionaries are not actually negative in financial contexts, leading them to create a finance-specific dictionary to improve sentiment classification. Simkin and Roychowdhury (2015) investigated the temporal decay of attention to news, suggesting that the timing of news articles affects the relevance of their sentiment signals for prediction. More recently, Araci (2019) addressed the problem of financial language directly by developing FinBERT, a version of BERT fine-tuned on financial texts. FinBERT outperformed several previous models in classifying financial sentiment, even with limited training data. The literature highlights the growing importance of domain-specific NLP tools and time-aware methods for understanding the relationship between news sentiment and financial markets, directly motivating our use of financial transformer models in this paper.

Data labeling is an important yet challenging step in supervised learning, often requiring significant time and resources. While this paper does not use labeled data directly for supervision, understanding labeling strategies remains valuable. The literature provides a wide range of approaches for data labeling in NLP and related domains. Yang et al. (2018) proposed using a dictionary to create distantly supervised labels and addressed two key issues, namely incomplete annotation and noisy labels, through partial annotation learning and reinforcement learning. Wang et al. (2023) tackled structural bias and label noise in distant supervision by designing a self-training framework that selects reliable pseudo-labeled samples and applies a debiasing module to improve token representations. Tang et al. (2023) focused on the analysis of the sentiment at the entity-level in financial texts. Although the work of Tang et al. (2023) does not directly propose a labeling framework, it demonstrates how entity recognition and sentiment models can be effectively combined in financial news analysis. Fredriksson et al. (2020) highlighted common labeling approaches such as crowdsourcing, where a large number of annotators contribute, and active learning, which iteratively selects the most informative samples for labeling. Both methods face challenges, including ensuring label quality, especially when domain knowledge is necessary. Semi-supervised learning offers a solution by training models on a small labeled set and then using them to label the remaining data, employing techniques like expectation-maximization and SVMs to improve results. In addition, Shelar et al. (2020) studied NLP tools for information extraction, focusing on NER. Their evaluation of libraries such as `spaCy`, `Apache OpenNLP`, and `TensorFlow` found `spaCy` to offer superior accuracy and speed, despite higher computational demands. This supports our choice of the `spaCy` transformer model in the proposed labeling framework.

1.2 Aim of the paper

This paper aims to improve financial text analysis by introducing a robust method for identifying stock associations and sentiment signals from large collections of domain-specific news articles. The main contribution is a novel ensemble labeling framework, FUNNEL (Filtered UNion for NER-based Ensemble Labeling) designed to improve the accuracy and coverage of article–stock associations. The proposed framework integrates weak keyword-based methods with `spaCy`'s transformer-based NER model, combining their strengths through a weighted voting mechanism. Label quality is further assessed through manual annotation using the financial news and stock price integration dataset (FNSPID). In addition, we derive sentiment signals using three pretrained models, namely, FinBERT, RoBERTa, and the valence-aware dictionary and sentiment reasoner (VADER), to generate time-series sentiment indicators.

The paper is organized as follows. Section 2 discusses the sentiment models. Section 3 introduces the dataset, outlining its structure, limitations, and issues related to labeling quality. Based on these observations, a labeling framework is proposed to improve label reliability. Section 4 presents the empirical results and a discussion of main results. Section 5 concludes the paper. Experiments on transformer-based stock price prediction are included in the supplementary material for illustrative purposes.

2 Sentiment analysis

We introduce the sentiment scoring models used to generate quantitative sentiment indicators from financial news articles. These scores serve as key inputs to stock price prediction models. We discuss three different approaches; two transformer-based models, namely FinBERT and RoBERTa, and a traditional rule-based method, VADER. We also explain how daily sentiment scores are aggregated and adjusted to align with financial data. Furthermore, to numerically represent the sentiment conveyed in each news article, we generate sentiment scores that typically range from −1 (negative) to 1 (positive), with 0 indicating neutrality. For our analysis, this scale is rescaled to [0, 5] to enable sentiment decay.

2.1 FinBERT

FinBERT is a modified variant of the BERT architecture specialized for financial NLP tasks, primarily targeting sentiment analysis to classify text as negative, neutral, or positive. While it shares the base BERT structure, FinBERT modifies the output layer depending on the task: a dense classification layer for classification tasks, and a regression output with mean squared error (MSE) loss for regression tasks. The main difference lies in FinBERT's fine-tuning strategy, designed to overcome catastrophic forgetting. It uses a slanted triangular learning rate schedule to shape the learning process and discriminative fine-tuning, which assigns progressively lower learning rates to lower layers of the model. Also, gradual unfreezing is applied, where initially only the final classification layer is trainable, and earlier layers are gradually unfrozen during training. FinBERT is pretrained and fine-tuned on three financial corpora, namely TRC2-financial, financial phrasebank and FiQA sentiment enabling it to better capture domain-specific vocabulary and nuances (Araci 2019).

2.2 RoBERTa

RoBERTa was developed as a replication aimed at creating a new pretrained language model based on the original BERT architecture. The quality and quantity of training data are crucial in developing a language model, as they significantly influence the model's vocabulary and the probability distributions underlying its predictions (Liu et al. 2019). To enhance performance, RoBERTa introduces several modifications to the original BERT training procedure. First, RoBERTa is trained on a substantially larger dataset, 160GB of uncompressed text drawn from five different corpora, compared to BERT's 16GB of uncompressed text. In addition, RoBERTa uses dynamic masking instead of static masking, generating new masking patterns during each training epoch. Finally, the training procedure involved longer sequences, larger batch sizes, and a greater number of training steps compared to BERT.

2.3 VADER

VADER is a rule-based approach sentiment analysis tool that assigns sentiment scores based on syntactic and grammatical heuristics. Each feature of the model, such as punctuation, capitalization, adverbs, conjunctions, emojis, acronyms, and slang, is associated with a valence score ranging from $[-4, 4]$. These heuristics include common features such as punctuation, capitalization, adverbs, and conjunctions, but also include western-style emojis or emoticons, acronyms, and slang. In total, the dictionary consists of over 7500 features that is scored within the previously mentioned range. When applied to a document, the text is tokenized and assigned a valence score if it exists in the dictionary. The negative, neutral, and positive scores are then summed and provide a sentiment score.

2.4 Sentiment averaging and decay

In this paper, we integrate both textual data (financial news) and numerical financial data (e.g., trading volume and prices), which have differing temporal resolutions. Direct concatenation based on the timestamps of news articles would likely introduce missing values due to sparse or irregular publication frequency. Conversely, aligning based solely on financial data timestamps could result in dimensional inconsistencies or data loss. To address these challenges, we compute daily average sentiment scores from financial news, thereby aligning a single sentiment value with the corresponding financial data for each day.

To prevent lookahead bias or information leakage, sentiment scores for each trading day t are computed using only news articles published up to and including that day. No information from future dates ($t + 1$, $t + 2$, ...) is used when constructing $Score_t$. This ensures that sentiment shows only information available to investors at that time and maintains temporal causality in the labeling, data fusion and forecasting steps.

Furthermore, in line with the reasoning of Simkin and Roychowdhury (2015), we recognize that sentiment is dynamic and unlikely to persist unchanged over time. For example, the sentiment expressed on day t may no longer be fully representative by day $t + 5$. To reflect this temporal decay, a smoothing adjustment is applied to daily sentiment scores that exceed an absolute value of 2.5, which are incrementally shifted toward the neutral baseline of 2.5, under the assumption that extreme sentiment naturally attenuates. The following function is applied as below

$$Score_t = 2.5 + \left(\frac{Score_{(t-1)} - 2.5}{t^{1.5}} \right), \tag{1}$$

where $Score_t$ gradually adjusts the previous day's sentiment score Score_{t-1} toward the neutral value of 2.5, scaled by a time-dependent decay factor $t^{1.5}$.

3 Data and labeling framework

We describe the datasets, preprocessing steps, and the novel ensemble-based labeling methodology. We address known limitations in financial text labeling by integrating weak supervision signals into a unified framework.

3.1 Data sources

Our primary source of financial news is the FNSPID introduced by Dong et al. (2024), which consists of two components: a textual component consisting of financial news articles, and a numerical component containing historical stock prices and

Table 1 Magnificent seven stock tickers

Stock	Stock ticker	Historical/alternative
Apple	AAPL	
Microsoft	MSFT	
Nvidia	NVDA	
Alphabet	GOOGL	GOOG
Amazon	AMZN	
Meta	META	FB
Tesla	TSLA	

trading volumes. Both parts are in time series format, covering the period from 1998 to 2024, and correspond to the same 4,775 S&P 500 companies. The two components are time-aligned. The textual part of the dataset includes 15.7 million news articles, organized by date and by the symbol of the corresponding stock.

To ensure relevance and high-frequency media coverage, we focus on the “Magnificent Seven” (MAG7) companies (Wang 2025): Apple, Microsoft, Nvidia, Alphabet, Amazon, Meta, and Tesla. This selection of MAG7 is supported by recent studies emphasizing their prominent roles in AI-driven market dynamics. For example, Basele et al. (2025) identified speculative bubbles in six of these seven stocks from December 2022 to January 2025, indicating heightened investor attention and volatility, while Umar et al. (2025) analyzed return and volatility spillovers among the seven largest technology companies. By concentrating on these firms, we use their substantial media presence and market influence, providing a robust foundation for evaluating our labeling framework. Table 1 lists the selected stocks and their associated ticker symbols.

3.2 Labeling challenges

While the FNSPID dataset[2] (Dong et al. 2024) offers a rich corpus of financial news articles; we observed substantial challenges related to labeling quality and temporal coverage. Initial labels, drawn from article links on NASDAQ’s stock-specific news feeds, were often incomplete or erroneous. Labeling inaccuracies reached up to 28%, with significant variability across stocks. For instance, the dataset for NVIDIA (NVDA) demonstrated long-term coverage and high accuracy, whereas Apple (AAPL) articles were mostly restricted to 2020–2023. However, a manual string search for “Apple” revealed articles from 2009–2024, suggesting many articles are

[2] The original dataset does not include manually annotated stock relevance labels. Instead, article-to-stock associations were inferred using a heuristic collection pipeline. Specifically, the original data collection process queried the NASDAQ website for news articles using a URL pattern of the form: https://www.nasdaq.com/market-activity/stocks/BLANK/news-headlines where `BLANK` was replaced with a stock ticker (e.g., AAPL, MSFT, TSLA, etc.). The resulting articles are then attributed to the corresponding stock and saved in stock-specific files.

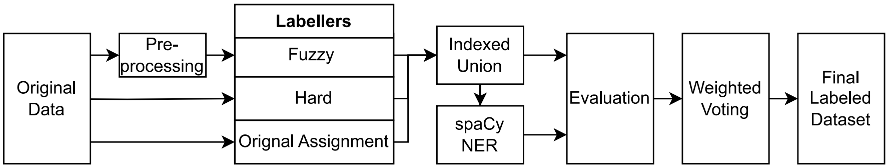

Fig. 1 Labeling methodology overview

missed due to a brittle labeling mechanism. These limitations motivated the development of an ensemble labeling pipeline (FUNNEL) designed to maximize recall while ensuring high precision. Additionally, FUNNEL allows a single article to be associated with multiple stocks when multiple tickers or company mentions are detected. This provides the realistic structure of financial news, where articles frequently discuss several firms simultaneously (e.g., "Big movers on NASDAQ today: AAPL, NVDA...").

3.2.1 Ensemble labeling approach (FUNNEL)

We developed the FUNNEL ensemble labeling framework, as illustrated in Fig. 1. As described above, this design allows articles to be linked to more than one stock, improving coverage over the original FNSPID assignments. Each component contributes to balancing recall and precision in the final ensemble labels. The ensemble incorporates multiple strategies, as detailed below:

1. *Original assignment*: The dataset's native *stock_symbol* field.
2. *Fuzzy matching*: Uses the `RapidFuzz` library to compute string similarity between pre-processed article titles and predefined stock-related keywords (threshold = 85)[3].
3. *Hard matching*: Exact string match against a refined keyword list.
4. *NER*: We applied spaCy's transformer-based NER model to identify mentions of target companies, it uses a predefined list to find PERSON, ORG, or PRODUCT entities corresponding to target companies.
5. *Application*: spaCy is only applied to articles flagged by at least one other labeling method, forming an "indexed union" subset. This design allowed fuzzy labeling to maximize recall, `spaCy` to ensure precision, and hard labeling to balance both.

3.2.2 Manual evaluation

We manually evaluated 1400 article-title–stock pairs across all labelers. For each stock, a random sample of 50 articles labeled by each method is reviewed. The evaluation measured labeling precision and informed ensemble weight selection.

[3] The threshold of 85 was empirically determined to balance recall and precision.

Let D_s be the set of all articles potentially related to stock s, and $N = |D_s|$. Suppose $\{x_1, ..., x_n\}$ be a random sample of size $n = 50$. The empirical accuracy of the k^{th} labeler is defined as

$$\hat{p}_k = \frac{1}{n} \sum_{i=1}^{n} y_i, \quad \text{where } y_i \in \{0, 1\},$$

with sampling variance and confidence interval

$$Var(\hat{p}_k) = \frac{\hat{p}_k(1 - \hat{p}_k)}{n} \left(\frac{N - n}{N - 1} \right), \quad \hat{p}_k \pm z_{\alpha/2} \sqrt{Var(\hat{p}_k)},$$

where $y_i = 1$ if the labeler's assignment for article i is correct, and 0 otherwise.

Table 2 shows `spaCy` NER achieved perfect precision (50/50) across all stocks, while fuzzy matching exhibited wide variance. The hard labeler is consistently strong (84–98%).

3.2.3 Weighted voting and ensemble model

We model the labels assigned by each labeler as independent Bernoulli random variables

$$y_i^{(k)} \sim \text{Bernoulli}(p_k).$$

The assumption of conditional independence underlies the maximum a posteriori interpretation of weighted voting and informs ensemble weight selection. We define a weighted vote function for each article i and stock s

$$score_i^s = \sum_{k=1}^{K} w_k y_i^{(k)}, \quad \sum_{k=1}^{K} w_k = 1,$$

with the final assignment

$$\hat{y}_i^s = \begin{cases} 1 & \text{if score}_i^s > \tau, \\ 0 & \text{otherwise.} \end{cases}$$

Table 2 Manual evaluation of labeling accuracy (correct labels out of 50)

Stock	Original	Fuzzy	Hard	spaCy
AAPL	37	29	45	50
MSFT	45	17	42	50
AMZN	36	35	43	50
GOOGL	37	36	48	50
NVDA	50	16	49	50
META	37	25	47	50
TSLA	42	26	48	50

where $y_i^{(k)} \in \{0, 1\}$ denotes the label assigned to article i by labeler k $(k = 1, \ldots, K)$, w_k is the weight reflecting confidence in the k-th labeler, the article is assigned to stock s if $score_i^s > \tau$, with threshold $\tau = 0.4$. The value of τ was empirically selected to balance precision and recall. Articles are included in the "indexed union" if flagged by at least one labeler, ensuring maximum coverage, while the high-precision NER component serves to filter potential false positives. The keyword-based labelers act as a high-recall filter for spaCy, which functions as a high-precision validator. Assuming conditional independence of labeler errors, the ensemble error corresponds to the probability that the weighted sum of votes does not exceed the threshold:

$$P(\text{error}_{\text{ensemble}}) = P\left(\sum_{k=1}^{K} w_k y_i^{(k)} \leq \tau \,\middle|\, L_i = 1 \right),$$

where L_i is the true latent label. While exact computation depends on the weights and threshold, this probability is generally lower than the error of any individual labeler, consistent with ensemble learning theory (Dietterich 2000), and can be estimated empirically using manually labeled data. Weighted voting can also be interpreted as a maximum a posteriori estimate over noisy labelers

$$P(L_i = 1 \mid y_i^{(1)}, \ldots, y_i^{(K)}) \propto \prod_{k=1}^{K} P(y_i^{(k)} \mid L_i = 1)^{w_k},$$

with thresholding at τ yielding the final assignment. From an information-theoretic perspective, each labeler contributes additional information, reducing the uncertainty (entropy) of the true label:

$$H(L_i \mid y_i^{(1)}, \ldots, y_i^{(K)}) < H(L_i \mid y_i^{(k)}) \quad \forall k.$$

The results of the manual evaluation support this theoretical motivation, showing that weighted ensembles improve overall precision and coverage. Figure 2 visualizes dataset size at each labeling stage.

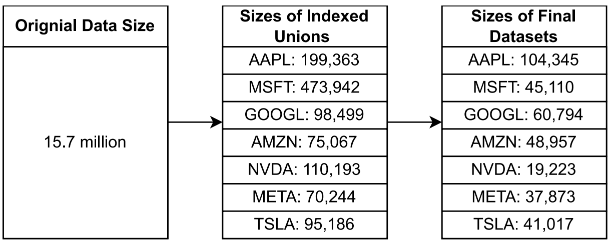

Fig. 2 Dataset size reduction at each stage of the labeling pipeline

3.3 Sentiment scoring and data fusion

After labeling, sentiment scores are computed using the article's combined title and body text. Each article is linked to a publication date and assigned to its respective stock. On days with multiple articles, the sentiment score is averaged. On days without news, the sentiment is decayed from the most recent prior update.

An example of the data structure post-merging with numerical data is shown in Table 3, where each row represents one day per stock with closing price, volume, and sentiment score.

Table 3 Example of final data structure per stock

Date	Volume	Closing price	Sentiment score	#Articles	Updated
2003-05-23	440502	213	2.5	23	1
2003-05-24	304000	211	2.121	0	0
2003-05-25	800023	201	1	203	1
⋮	⋮	⋮	⋮	⋮	⋮

Table 4 Descriptive statistics of dates and stock prices

Stock	Date range		Closing price			
	Start	End	Mean	Std	Min	Max
AAPL	2006-01-03	2023-12-26	47.475	53.972	1.810	198.110
GOOGL	2005-05-02	2023-12-26	43.266	38.606	5.560	149.840
META	2012-05-18	2023-12-26	157.490	90.749	17.730	382.180
MSFT	2005-04-28	2023-12-26	92.849	95.967	15.150	382.700
TSLA	2010-06-29	2023-12-26	70.312	101.379	1.050	409.970
NVDA	2005-04-28	2023-12-26	5.546	9.824	0.150	50.410
AMZN	2005-04-28	2023-12-26	48.166	54.082	1.300	186.570

Table 5 Descriptive statistics of trading volumes

Stock	Trading volume			
	Mean	Std	Min	Max
AAPL	385858338.536	387643101.747	24048344	3373059549
GOOGL	64431068.913	62716598.959	9312760	823657780
META	30645940.421	26447841.865	5467488	580587742
MSFT	45123561.635	28405226.355	7425603	591078581
TSLA	96876509.389	80144525.730	1779212	914082284
NVDA	550569387.311	324669522.568	45645120	3692927840
AMZN	107450731.335	90032640.443	17626740	2088091780

Table 6 Descriptive statistics of sentiment scores by model and stock

Stock	FinBERT		RoBERTa		VADER	
	Mean	Std	Mean	Std	Mean	Std
AAPL	4.167	0.834	2.836	0.528	3.561	0.935
GOOGL	4.215	1.022	2.793	0.636	3.477	1.034
META	3.972	1.231	2.877	0.827	3.705	1.139
MSFT	4.074	1.096	2.770	0.628	3.727	1.080
TSLA	4.052	1.105	2.795	0.684	3.660	1.139
NVDA	3.377	1.291	2.724	0.680	3.417	1.274
AMZN	3.966	1.127	2.869	0.694	3.800	1.095

Table 7 Daily article count statistics and sentiment update proportion per stock

	Article count statistics				Sentiment updated		Total
	Mean	Std	Min	Max	True	False	Articles
AAPL	21.509	27.612	0	816	0.950	0.050	97 348
GOOGL	12.075	26.553	0	779	0.921	0.079	56 703
META	9.029	18.723	0	369	0.921	0.079	26 364
MSFT	9.065	12.968	0	372	0.890	0.110	42 584
TSLA	10.618	26.203	0	598	0.925	0.075	36 069
NVDA	3.738	17.092	0	430	0.469	0.531	17 560
AMZN	9.536	21.684	0	596	0.858	0.142	44 798

4 Experiment results and discussions

4.1 Summary statistics

We provide the summary statistics in Tables 4, 5, 6 and 7 of the core variables used in our modeling pipeline. Table 4 reports the date range for each stock, where the start date is determined by the availability of closing price data. The dataset covers different time spans across stocks due to varying IPO dates and data availability. For example, META (formerly Facebook) data starts in 2012, while MSFT and NVDA data begin in 2005. The mean closing prices reflect differences in valuation levels, with META, MSFT, and TSLA showing relatively high standard deviations, suggesting elevated volatility over the observed period. These variations are important for model calibration, especially when using price-normalization techniques.

Table 5 presents average and extreme values of daily trading volumes. The sheer scale of volume varies greatly, NVDA and AAPL exhibit especially high average volumes, often exceeding hundreds of millions, while GOOGL and META are more modest. NVDA's maximum volume is particularly noteworthy (over 3.6 billion), driven by speculative trading in certain periods. Table 6 shows the mean and standard deviation of sentiment scores derived from FinBERT, RoBERTa, and VADER.

FinBERT consistently produces more optimistic sentiment values (mean scores > 3.9), while RoBERTa is more neutral (means around 2.7 to 2.8). VADER occupies a middle ground. Notably, FinBERT also exhibits higher standard deviation, indicating it differentiates more strongly between positive and negative news than the other models. Finally, Table 7 provides statistics on the number of articles per stock, including the daily distribution and the percentage of days where sentiment scores were updated. AAPL leads in total news volume, averaging 21.5 articles per day and peaking at 816 on a single day. On the other hand, NVDA has the lowest average and update rate, only 47% of days had sentiment updates, making it a natural candidate for robustness testing. These descriptive statistics confirm the heterogeneity of our dataset across both textual and numerical dimensions.

4.2 Labeling application

The fuzzy labeler was constructed using the `RapidFuzz` Python package (Bachmann 2025), where string similarities between article titles and a predefined list of relevant keywords were considered a match if the partial ratio scorer (PRS) returned a similarity score above 85. This threshold was selected empirically after preliminary experiments across multiple stocks, where lower cut-offs introduced excessive noise while higher cut-offs reduced recall. The PRS attempts to find the optimal alignment between two strings and returns a similarity score as a float between 0 and 100. As the scorer is sensitive to casing and word order and can be affected by such things as punctuation, preprocessing was recommended by the package developers. Thus, before applying the function, the data was preprocessed by removing punctuations, forcing lowercase and removing common English stopwords, as described in Marcucci (2024, 425-468). The preprocessing was conducted using spaCy's `en_core_web_sm` model and took 6:38:35 on an Apple Intel 2020 CPU for the entire corpus of 15,549,298 article titles. The same spaCy model was used for NER labeling to identify relevant stocks by relating recognized entities (PERSON, ORG, or PRODUCT) to a predefined list of companies. For this task, a NVIDIA A100

Table 8 Computation time per stock and labeler

Stock	Computation time		
	Fuzzy	Hard	spaCy NER
AAPL	22:22	01:35	06:51
GOOGL	19:38	01:20	04:27
META	14:37	01:05	02:36
MSFT	25:26	01:14	15:37
TSLA	17:12	01:14	03:55
NVDA	14:45	01:04	04:45
AMZN	17:09	01:15	03:07

Time in MM:SS. Fuzzy and Hard used Apple Intel 2020 CPU, spaCy NER used NVIDIA A100 GPU

GPU was utilized via Google Colab's cloud computation service. Table 8 reports the actual computation times.

Appendix Figs. 3, 4, 5, 6, 7, 8 and 9 compare article volumes across labeling methods for each stock. The plots illustrate how different strategies contribute: keyword-based methods (fuzzy and hard matching) expand coverage substantially, while spaCy NER applies targeted refinements within their union. The final ensemble (black line) captures both the breadth of coverage and the precision of transformer-based recognition, producing a more stable and continuous labeling outcome than the original FNSPID labels (red), which display noticeable gaps and sudden shifts. Where keyword-based models agreed, they dominated; otherwise, spaCy determined inclusion.

The figures also highlight the computational efficiency of this design. For example, in META (Fig. 6), the fuzzy labeler identifies a large pool of candidate articles, but many spurious matches are filtered out once spaCy NER is applied. Instead of running NER across the entire corpus of 15.7 million articles, it is restricted to this reduced set, which markedly lowers computational cost. Similarly, in MSFT (Fig. 7), the ensemble corrects abrupt jumps seen in the original labels by combining the high recall of keyword methods with the high precision of spaCy. This division of labor between fast, low-cost models and selective NER ensures that recall is not sacrificed for precision, while keeping GPU-intensive computations manageable. Appendix Figs. 3, 4, 5, 6, 7, 8 and 9 demonstrate that the FUNNEL framework achieves both accuracy and efficiency, providing a practical framework for scaling financial news labeling to large datasets.

4.3 Discussion

The results highlight both the strengths and the limitations of the FUNNEL. First, the issue of label quality proved central. Manual evaluation showed that the original FNSPID labels were often incomplete or inaccurate, with error rates approaching 28% for certain stocks. In contrast, the ensemble strategy improved reliability, with spaCy NER achieving perfect precision in the manual sample and hard keyword matching providing consistently high accuracy. Although the fuzzy labeler exhibited more variability, it contributed valuable recall by capturing articles missed by stricter methods. The weighted voting scheme balanced these characteristics, producing a labeled dataset that was more continuous and stable than the original. This suggests that our approach is capable of addressing common weaknesses in large-scale financial text corpora, where both over- and under-labeling are frequent problems.

Second, the computational effort required to implement the pipeline was nontrivial. Preprocessing and fuzzy matching over the 15.7 million article titles required nearly seven hours on a CPU, while NER labeling with spaCy demanded access to a high-performance GPU. Although these requirements are substantial, the trade-off is a significant improvement in label reliability and robustness. Moreover, the design of the pipeline allowed efficient division of labor between methods: keyword-based models provided high recall at relatively low cost, while NER acted as a precision filter applied only to a reduced subset. This modular structure demonstrates that improved quality can be achieved without prohibitive computational expense, provided resources are allocated strategically.

Third, the ensemble model itself illustrates the importance of integrating labeling strategies. By defining a weighted vote threshold, the framework ensured that no single method dominated the labeling process, while still allowing spaCy to act as a decisive validator in ambiguous cases. This design addresses the tension between coverage and accuracy, which is particularly acute in financial text applications where context-specific terminology complicates naive keyword approaches. The outcome is a pipeline that generalizes across stocks with heterogeneous levels of media coverage and data quality.

Finally, the analysis of sentiment signals underscores their role in shaping downstream financial tasks. Systematic differences between models were evident, with FinBERT producing more optimistic and more variable scores, RoBERTa clustering around neutrality, and VADER occupying an intermediate position. These polarity distributions highlight the need for caution when selecting sentiment inputs, as model-specific biases may influence empirical findings. A more detailed evaluation of how these sentiment signals interact with forecasting architectures is presented in the supplementary material.

5 Conclusion and future work

This paper introduces an ensemble-based framework for improving sentiment inputs in financial modeling. The primary contribution is the FUNNEL labeling approach, which combines keyword heuristics and transformer-based NER into a weighted voting system. Manual evaluation demonstrates that this framework improves label reliability relative to the original dataset, while analysis of sentiment scores reveals systematic differences among FinBERT, RoBERTa, and VADER. A case study, presented in the supplementary material, further illustrates how these sentiment signals can be employed in time series prediction. In summary, the results indicate that FUNNEL provides a reliable and broadly applicable framework for incorporating sentiment information into financial datasets, provided that preprocessing and label quality are carefully managed. Despite these findings, the study has some limitations. The focus of this study was restricted to MAG7 companies, which limits the breadth of empirical generalization. Sentiment was derived from three pretrained models without additional fine-tuning, and model-specific polarity biases may have affected the results. The computational demands of preprocessing and labeling remain considerable, although the modular design of the pipeline helps to mitigate this issue.

Future research could investigate the effect of sentiment decay across different prediction horizons, test alternative model architectures, or expand the dataset to include more diverse equities. Incorporating classical baselines and investigating the fairness, environmental impact, and computational efficiency of the model would also be valuable. Conducting ablation studies of the FUNNEL pipeline represents an important direction for future research, allowing the relative impact of fuzzy, hard, and NER components on label quality to be assessed. Extending sentiment modeling to include additional or fine-tuned models could help mitigate polarity bias and assess robustness. Another natural extension is to explore the application of FUNNEL-labeled sentiment scores in portfolio optimization or stock selection strategies, thereby evaluating the practical utility of the framework in systematic investment analysis. Collectively, these directions emphasize FUNNEL's potential as a general-purpose framework for reliable financial sentiment labeling.

Labeling visualizations

See Appendix Figs. 3, 4, 5, 6, 7, 8 and 9

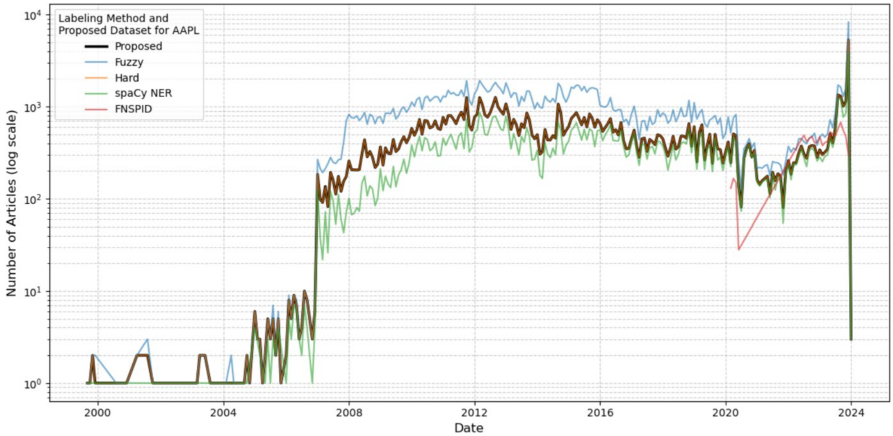

Fig. 3 Comparison of labeling methods for AAPL

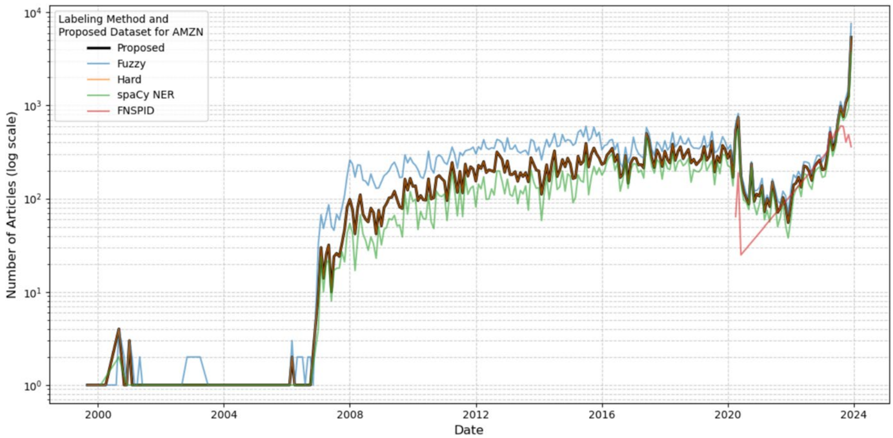

Fig. 4 Comparison of labeling methods for AMZN

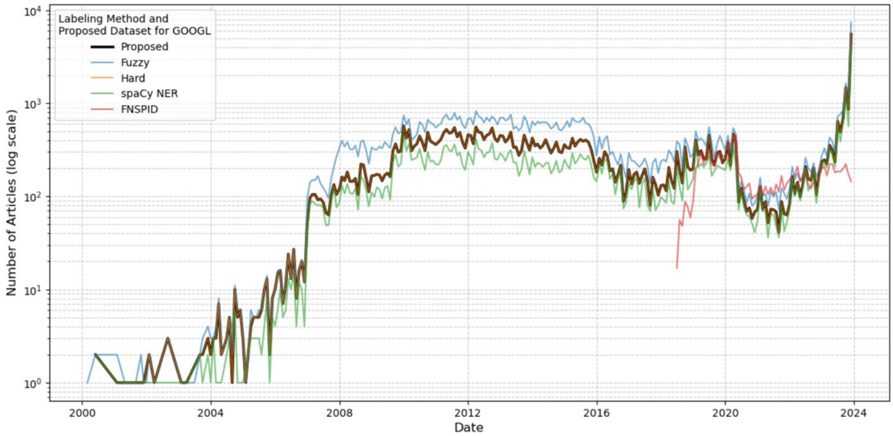

Fig. 5 Comparison of labeling methods for GOOGL

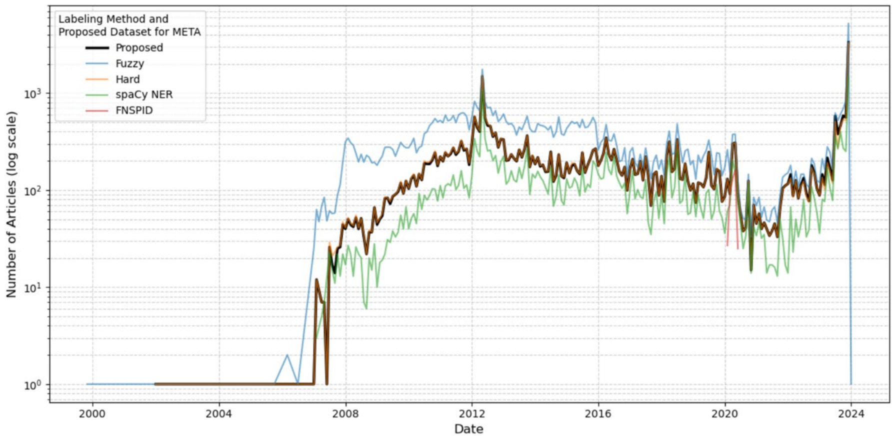

Fig. 6 Comparison of labeling methods for META

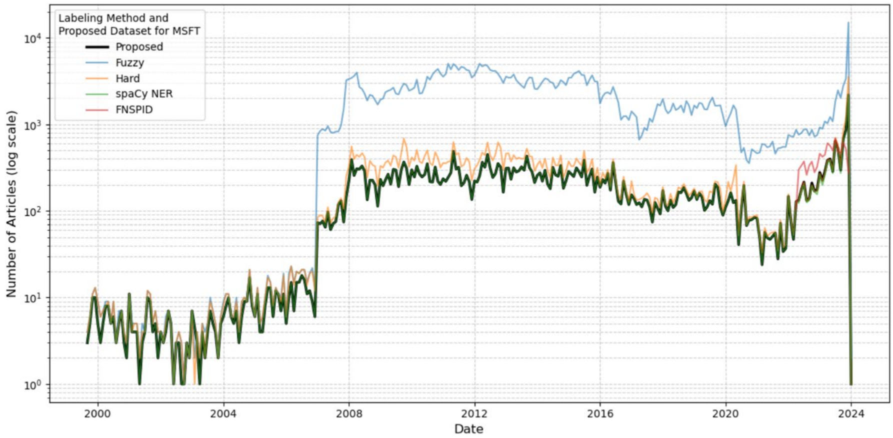

Fig. 7 Comparison of labeling methods for MSFT

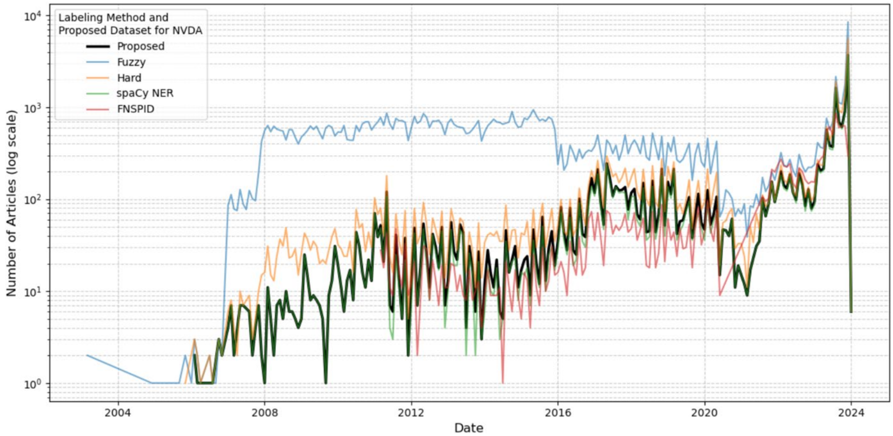

Fig. 8 Comparison of labeling methods for NVDA

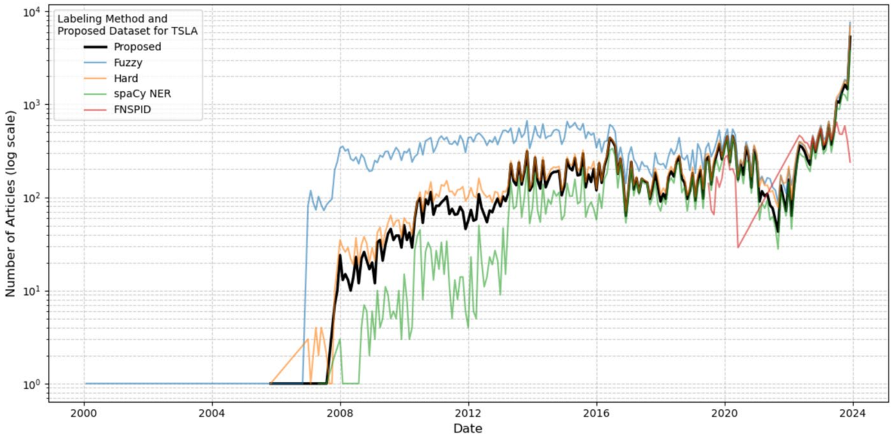

Fig. 9 Comparison of labeling methods for TSLA

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Author contributions WN, FF, and MQ conceptualized the study and designed the methodology. WN and FF developed the code, conducted the experiments, and analyzed the results. MQ supervised the project and drafted the manuscript. All authors (WN, FF, MQ) reviewed, edited, and approved the final version of the manuscript.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interest The authors declare no competing interests.

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